Global critical materials

SUPPLY INTELLIGENCE / EUROPE

DESIGN PREVIEW | DESKTOP QA PENDING

Where are mining and refining concentrated?

STEPS / Base Case | Mining in forest green; Refining in leaf green | Each stage has its own panel denominator

EXACT MATERIAL DEFINITION

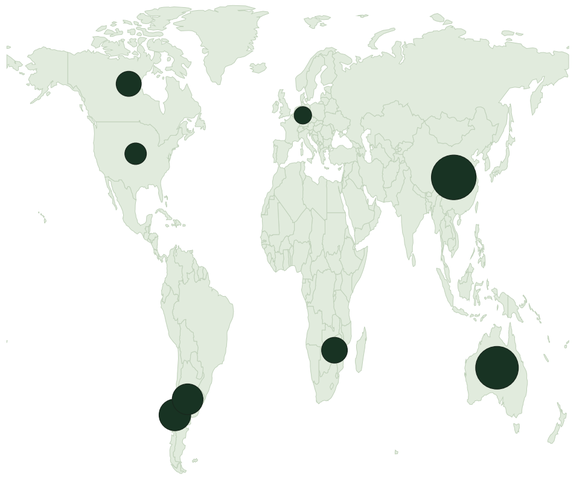
C | All grades | C | Graphite | Co | Cobalt | Cu | Copper | **Li | Lithium** | Ni | Nickel | REE | Magnets

SOURCE YEAR

2025 | **2030** | 2035 | 2040

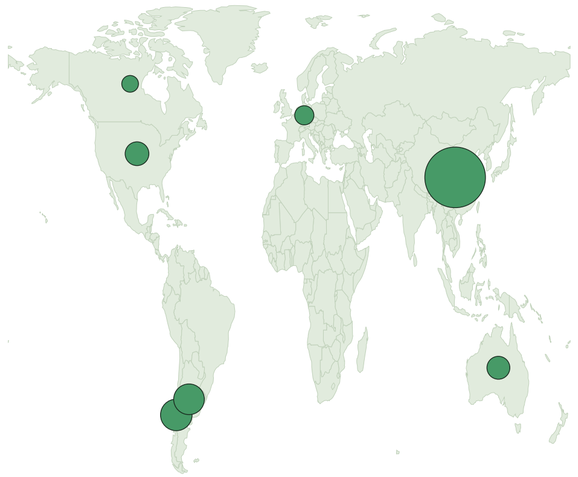
Mining | Expected output	Refining | Expected output	Mining | HHI bounds	Refining | HHI bounds	Mining | Top 3 share	Refining | Top 3 share
540.2 kt	543.7 kt	0.185 - 0.198	0.483 - 0.485	66.9% - 67.8%	87.8% - 87.8%

Mining | named-country share



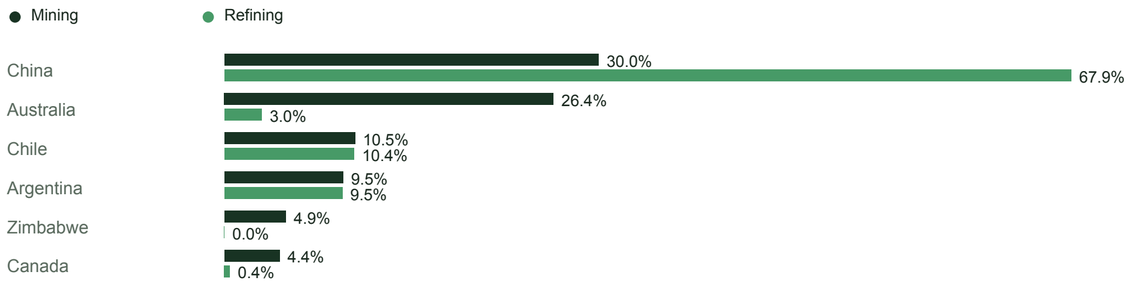
Named countries only | Natural Earth preview

Refining | named-country share



Named countries only | Natural Earth preview

Country shares | Mining / Refining



Top 6 of union of named countries shown in preview; native chart scrolls.

Expected output | Mining / Refining | native kt



Expected output, not capacity. 2025 baseline / native basis unverified. Other countries is opaque. All-grades graphite supply was not returned; blanks stay blank.

Global critical materials

SUPPLY INTELLIGENCE / EUROPE

Global Supply Landscape | **European Trade & Dependency** | Supply Chain Risk | Future Outlook

DESIGN PREVIEW | DESKTOP QA PENDING

How exposed is the selected reporting area?

Countries: all partners, including intra-EU | EU27: external partners only, fixed 2020 membership | Select one reporting area

REPORTING AREA | EU27 IS THE BLOC, NOT THE SUM OF THESE SIX COUNTRIES

EU27 | Extra-EU | France | **Germany** | Italy | Netherlands | Poland | Spain

TRADED PRODUCT | NATIVE CN8

Co oxides | 28220000 | Cu ores | 26030000 | Graphite | 25041000 | **Li carbonate | 28369100** | Ni ores | 26040000 | REE magnets | 85051110

SNAPSHOT YEAR

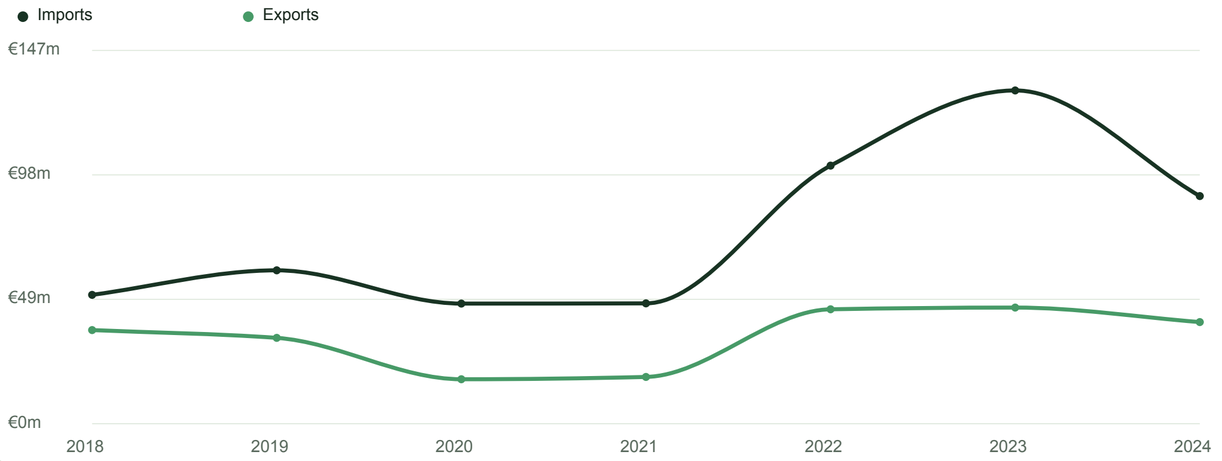
2018 | 2019 | 2020 | 2021 | 2022 | 2023 | **2024**

MEASURE BASIS

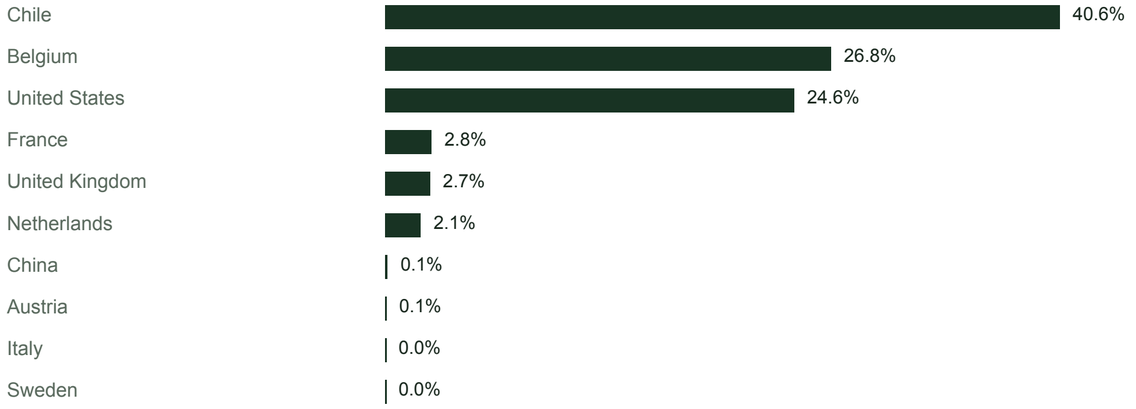
Product mass | t | **Value | EUR**

Imports | Selected scope	Exports | Selected scope	Trade balance	Top supplier share	Supplier HHI
€89.7m	€40.0m	€-49.6m	40.6%	0.300

Imports / Exports | selected reporting scope



Import partners | share of selected scope total



Dependency unavailable: consumption / domestic output missing. HHI needs a positive import total. Native CN8 continuity unverified. CIF / FOB; partner is not mine origin. German cobalt mass discrepancy unresolved.

SOURCE: IEA | GCMO2026 / EUROSTAT | DS-045409 | IEA: 30 OCT 2026 / TRADE: 30 OCT 2026-01 NOV 2026 | PORTFOLIO ANALYSIS

Global critical materials

SUPPLY INTELLIGENCE / EUROPE

DESIGN PREVIEW | DESKTOP QA PENDING

Where does concentration sit along the chain?

Above the diagonal: refining lower-bound HHI exceeds mining | Below: mining exceeds refining | This compares concentration, not physical deficits

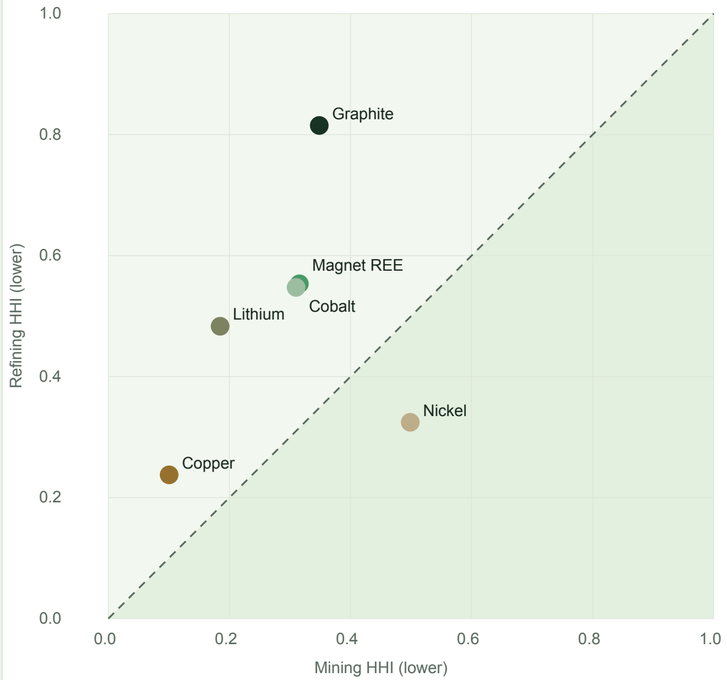
NATIVE DEFINITION | OPTIONAL FILTER

C | All grades | C | Graphite | Co | Cobalt | Cu | Copper | Li | Lithium | Ni | Nickel | REE | Magnets

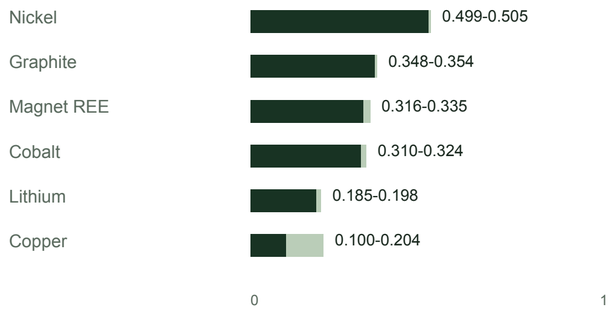
SUPPLY REFERENCE YEAR

2025 | **2030** | 2035 | 2040

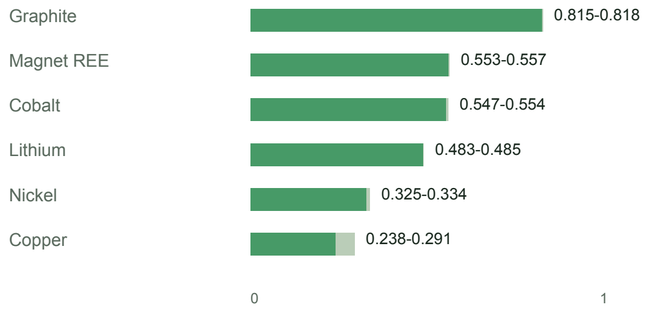
Mining vs. Refining | HHI lower bounds



Mining concentration | HHI bounds



Refining concentration | HHI bounds



TRADE REPORTING AREA

Germany ∨

TRADE SNAPSHOT YEAR

2018 | 2019 | 2020 | 2021 | 2022 | 2023 | **2024**

Selected area supplier HHI | Value / Mass

Product	Value HHI lower	Mass HHI lower
Cobalt oxides / hydroxides	0.706	0.683
Copper ores / concentrates	0.189	0.188
Lithium carbonates	0.300	0.290
Natural graphite powder / flakes	0.166	0.199
Nickel ores / concentrates	0.369	0.379
Rare-earth-containing magnets	0.751	0.754

Diagonal boundary = equal lower-bound HHI. Full bounds remain in tooltips. Coloured bar = lower bound; muted extension = upper − lower.

HHI base + interval width = upper bound; not a statistical boxplot. Import dependency, physical gap and capacity unavailable: compatible inputs absent. German cobalt mass discrepancy unresolved.

SOURCE: IEA | GCMO2026 / EUROSTAT | DS-045409 | IEA: 30 OCT 2026 / TRADE: 30 OCT 2026-01 NOV 2026 | PORTFOLIO ANALYSIS

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How does the outlook change across scenarios?

Native 2025 baseline and scenario years | Demand sums returned end uses | Supply is available only in STEPS

EXACT MATERIAL DEFINITION

C | All grades | C | Graphite | Co | Cobalt | Cu | Copper | **Li | Lithium** | Ni | Nickel | REE | Magnets

KPI SCENARIO

CPS | HDS | **STEPS**

KPI ENDPOINT

2025 | 2030 | 2035 | **2040** | 2045 | 2050

Demand at endpoint | native scope

932.5 kt

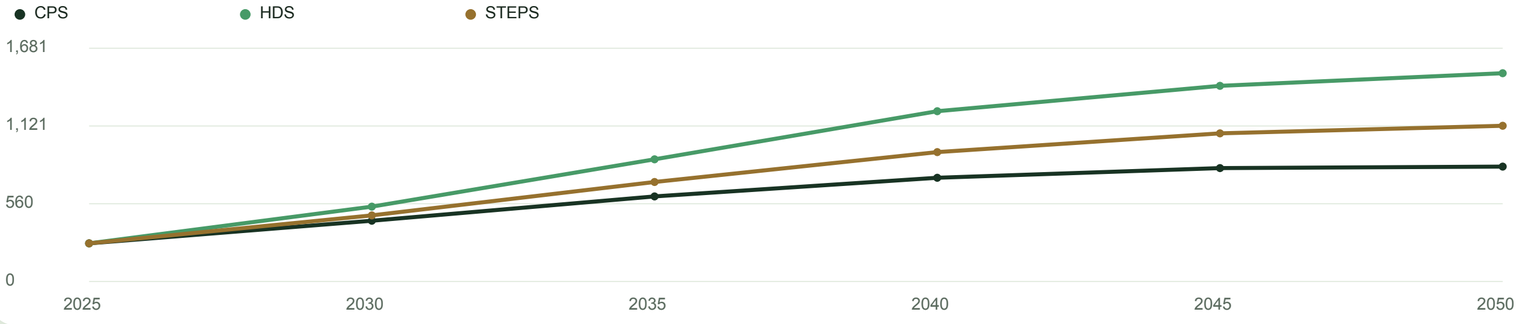
Demand growth from baseline

+239.1%

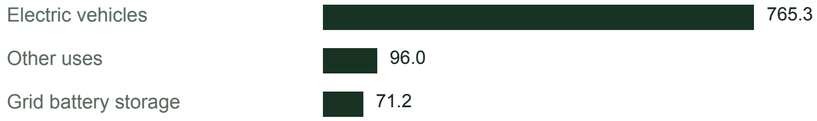
Annualised growth

8.5%

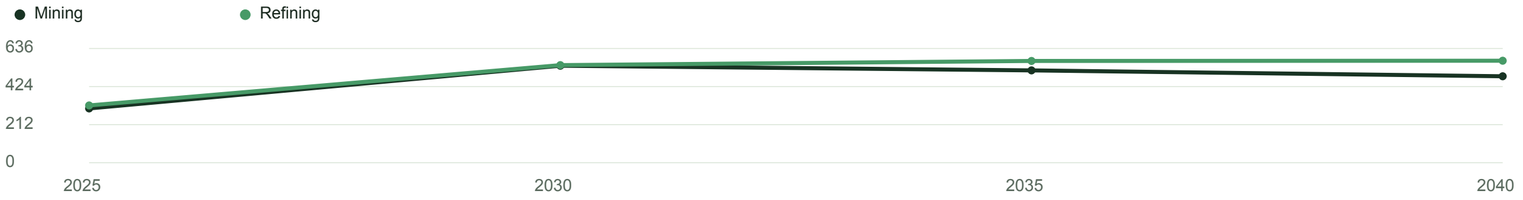
Demand scenarios | native kt | chart keeps all scenarios



End uses | selected KPI scenario / endpoint | kt



Expected output | Mining / Refining | STEPS | native kt



Physical gap unavailable: matched primary requirements / native basis unverified. Output is not capacity. Supply ends in 2040; demand extends to 2050. Connecting lines are guides.

SOURCE: IEA | GCMO2026 / EUROSTAT | DS-045409 | IEA: 30 OCT 2026 / TRADE: 30 OCT 2026-01 NOV 2026 | PORTFOLIO ANALYSIS